

QUESTION 1

You are structuring a table relational database.

For each statement, select **True** or **False**.

Note : you will receive partial credit for each correct selection.

	True	False
Each value in a field in a table must be unique (F)	☐	☒
Each row in a table must be unique (F)	☒	☐
Each column name in a table must be unique (T)	☒	☐

QUESTION 2

You have a table named **Product**. The product table has column for **productDescription** and **ProductCategory**.

You need to change the **ProductCategory** value for all the spoons in the **Product** to 43.

A **ProductDescription** of spoon indicates that the item is a **spoon**.

Which statement should you use?

- A. UPDATE Product
WHERE ProductDescription = 'spoon'
SET ProductCategory = 43
- B. UPDATE Product
SET ProductCategory = 43
WHERE ProductDescription = 'spoon'
- C. UPDATE Product
WHERE ProductDescription = 'spoon'
TO ProductCategory = 43
- D. UPDATE Product
TO ProductCategory = 43
WHERE ProductDescription = 'spoon'

QUESTION 3

You accept an IT internship at a local charity. The charity ask you to keep a record of its volunteers by using a database table named **Volunteer**. The table has the following columns and rows:

ID	GivenName	DateOfBirth
1	Tia	1976-05-30
2	Susana	1952-11-04
3	Joey	1963-02-17

You must update the table when volunteers ask to be removed from mailing list

You need to **delete** all records with the **GivenName** Tia

Which SQL statement should you use ?

- A. DELETE FROM Volunteer WHERE GivenName = 'Tia'
- B. DELETE FROM Volunteer WHERE GivenName = 'Tia'

- C. DELETE FROM Volunteer WHERE GivenName EQUALS 'Tia'
D. DELETE FROM Volunteer WHERE GivenName IS 'Tia'

QUESTION 4

You delete row in a table named **Order**. The corresponding row in the **OrderItem** table are automatically deleted

This process is an example of a/an :

- A. Domino delete
B. Waterfall delete
C. Functional delete
D. Cascade delete
E. Inherited delete

QUESTION 5

A table named PlayerStat contain following fields :

Field	Data Type	Allow Nulls
PlayerID	INT	FALSE
TeamID	INT	FALSE
GameDate	DATETIME	TRUE
Point	INT	TRUE

You need to display the total number of **point** per player on the **team** whose ID is 1.

Move the appropriate code segments from the list on the left to the correct locations on the right. You may use each code segment once, more than once, or not at all.

Note : you will receive partial credit for each correct selection.

The screenshot shows a SQL query builder interface. On the left, under 'Code Segments', there are several draggable boxes: 'PlayerID', 'TeamID', 'SUM(Point)', 'COUNT', 'WHERE', 'HAVING', and 'GROUP BY'. On the right, under 'Answer Area', the assembled query is shown: 'SELECT PlayerID, SUM(Point) FROM PlayerStat WHERE TeamID = 1 Group By PlayerID'. The query is built using the segments from the left.

QUESTION 6

You create the following query to determine whether Sample Movie appears only once in the Movie table.

```
SELECT Title
FROM Movie
WHERE Title = 'Sample Movie'
ORDER BY Title
GROUP BY Title
```

HAVING COUNT(*) = 1

When you run this Query, it returns a syntax error

You need to modify the query to run without error and return accurate result

What should you do?

- A. Remove the group by clause
- B. Change the having clause to having count(title) = 1
- C. Remove the order by clause
- D. Change the having clause to having count(1) = 1

QUESTION 7

Which statement creates an index ?

- A. CREATE TABLE Employee
(EmployeeID INTEGER DISTINCT)
- B. CREATE TABLE Employee
(EmployeeID INTEGER INDEX)
- C. CREATE TABLE Employee
(EmployeeID INTEGER PRIMARY KEY)
- D. CREATE TABLE Employee
(EmployeeID INTEGER NULL)

QUESTION 8

You have the following table definition.

Create Table Road
(Roadid Integer Not Null,
Distance Integer Not Null)

The road table contains the following data.

RoadID	Distance
1234	22
1384	34

You execute the following statement.

INSERT INTO Road VALUES (1234, 36)

What is the result?

- A. A new row in table
- B. A syntax error
- C. An error stating that null values are not allowed
- D. An error stating that duplicate ids are not allowed

QUESTION 9

A stored procedure contains the following query

SELECT 'Greetings' + Prefix + ' ' + Firstname FROM Person;

The stored procedure returns all null values. You verify that there is data in the person table

What is likely the cause of the problem

- A. You must specify the NULLIF keyword in the query
- B. The plus (+) operator cannot be used to append character data
- C. The Prefix Of FirstName columns have null values
- D. You must specify the join keyword in the select statement

QUESTION 10

You are writing a **SELECT** statement to find every product whose name contains a **specific** character.

Which keyword should you use in your **WHERE** clause.

- A. Like
- B. In
- C. Between
- D. Fetch

QUESTION 11

Which feature does a relational database use to ensure that data entered into a column is valid?

- a. An attribute
- b. A primary key
- c. A constraint
- d. An index

QUESTION 12

The customers table includes the following data

ID	FirstName	Lastname	PhoneNumber	Extension
1	Hope	Ragabash	(123)555-0111	NULL
2	Luna	Faltor	(123)555-0112	NULL
3	Mickey	Sassafras	(123)555-0113	12
4	Minne	Hemingway	(123)555-0114	77
5	Sherlock	Steam	(123)555-0115	NULL

You need to create a query that returns a result set that contains the lastname, phonenumbr and extension that are valid numbers. The result set should be sorted by the customer's last name.

Complete the code by selecting the correct option from each drop-down list **Order BY**

Answer Area

```
SELECT LastName, PhoneNumber, Extension
FROM Customers
WHERE Extension IS NOT NULL
GROUP BY LastName;
```

QUESTION 13

You have a table product that contains one million row. You need to search for product information using the following query :

SELECT ProductName, Price FROM Product WHERE Category = ' Science Books'

What will make this type of search more efficient?

- A. A clustered index on the ProductName column
- B. A clustered index on the Price column
- C. A non-clustered index on the Category column**
- D. A non-clustered on the Price column

QUESTION 14

You have a Customer table that store information about customers and an Order table that stores information about orders placed. Each table contains a CustomerID column. You join the two tables by using the CustomerID column in each table.

You run a query the selects the following data.

- All customers and their orders
- Costumers who have no orders

Which type of join do these results represent ?

- A. Full join**
- B. Partial join
- C. Inner join
- D. Complete join

QUESTION 15

You need to create a report of data in the students table. The report must meet the following requirements:

- Display all student who enrolled on or after June 1, 2020
- Display all student who graduated in 2020
- Return the result set in order of enrollment, with the most recent enrollment date first

Enrollment_Date specifies the enrollment date, Graduation_Date specifies the graduation date.

An Academic_Status Of Graduated indicates that a student has graduated.

Which query should you use ?

- A. SELECT * FROM student
WHERE enrollment_date >= '2020-06-01' OR academic_status ='graduated' AND
graduation_date >= '2020-01-01'
ORDER BY enrollment_date DESC**
- B. SELECT * FROM student
WHERE enrollment_date >= '2020-06-01' AND academic_status ='graduated' AND
graduation_date >= '2020-01-01'
ORDER BY enrollment_date ASC
- C. SELECT * FROM student
Where enrollment_date >= '2020-06-01' AND academic_status ='graduated' OR
graduation_date >= '2020-01-01'

```

ORDER BY enrollment_date
D. SELECT * FROM student
WHERE enrollment_date >= '2020-06-01' OR academic_status = 'graduated' OR
graduation_date >= '2020-01-01'
ORDER BY enrollment_date DESC

```

QUESTION 16

Which statement should you use to remove a foreign key?

- a. Delete table
- b. Alter table**
- c. Alter foreign key
- d. Delete foreign key

QUESTION 17

You accept an IT internship at a local charity. The charity has two tables in their data model named chapter and language, as shown

Chapter table

ChapterID	City	Region	Country
1	Chicago	Illinois	USA
2	Los Angeles	California	USA
3	New York City	New York	USA
4	Toronto	Ontario	Canada

Language table

LanguageID	LanguageName	Locale
1	English	USA
2	English	Canada
3	Spanish	Spain
4	Spanish	Mexico
5	French	France
5	French	Canada
6	German	Germany

You create a third table named ChapterLanguage to relate the chapter table and the Language Table.

You need to select columns from the Chapter and Language tables to create a composite primary key for the ChapterLanguage table.

Which two columns should you select ? (select 2)

- A. Chapterid**
- B. City
- C. Region
- D. LanguageName
- E. Country
- F. Languageid**

QUESTION 18

You write a series of SQL statements that perform a long, complex query. The statements then send the result in an email.

You want to be able to manually call the code any time you need to.

Which database object can you use to store the code ?

- A. Stored procedure
- B. Function
- C. Trigger
- D. View

QUESTION 19

The products table contains the following data.

ItemNumber	ItemName	ItemDescription	Price
1	Bonbon Box	Chocolate Truffles, Black Forest Truffles, and Cherry Bonbons	24.95
2	Brownie Mini Bites	Caramel Nut, German Chocolate, Chocolate Swirl, and Double Dutch Brownies	14.95
3	Cappuccino Companion Collection	Tasty treats to accompany your favorite java, Includes Strawberry Thumbprints, Chocoate Chip Cookies, and Biscotti.	21.50
4	Citrus Cooler Gift Box	Refreshing citrus cookies, including Orange Chiffon Meltaways, Key Lime Bars, Lemon Coolers, and Tangerine Coconut Smoothies	19.99
5	Fruit Jewels Assortment	Fruity Favorites, including Apricot Jewels, Cherry Bonbons, Coconut Patties, and Strawberry Thumbprints	29.99

Which query will increase the price of item 1 by 6 percent ?

- A. USE Products
SET Price = Price * 1.06
WHERE ItemNumber = 1;
- B. SET Price = Price * 1.06
FROM Products
WHERE ItemNumber = 1;
- C. ALTER Products
SET Price = Price * 1.06
WHERE ItemNumber = 1;
- D. UPDATE Products
SET Price = Price * 1.06
WHERE ItemNumber = 1;

QUESTION 20

You need to delete a database table.

Which data definition language (DDL) keyword should you use ?

- A. Alter
- B. Delete
- C. Drop
- D. Truncate

QUESTION 21

You execute the following query.

```
SELECT EmployeeID, FirstName, DepartmentName  
FROM Employee, Department
```

This type of operation is called a/an

- A. Intersection
- B. Outer join
- C. Cartesian product
- D. Equi-join

QUESTION 22

You need to normalized a database to first normal form.

Which two requirement must you meet? (Choose 2)

- A. Exclude repeating groups
- B. Exclude foreign key
- C. Exclude composite key
- D. Exclude duplicate rows

QUESTION 23

A database table stores information about school attendance. You have a data set that is defined as follows :

StudentName	GradeLevel	DaysAbsent
John	12	2.5
Holly	12	0.0
David	12	3.0

You have the following requirements

- Studentname must consist of a string of characters
- Gradelevel must be whole number
- Dayabsent can have one number after the decimal

Move each appropriate data type from the list on the left to the correct column name on the right. You may use each data type once, more than once, or not at all.

Data Types		Column Names
BIT	DATETIME	StudentName
DECIMAL	INT	GradeLevel
VARCHAR	CHAR	DaysAbsent

QUESTION 24

Which statement deletes the rows where the employee's phone number is not entered

- A. DELETE FROM employee
WHERE phone = NULL
- B. DELETE FROM employee
WHERE phone IS NULL
- C. DELETE FROM employee
WHERE phone IS NOT NULL
- D. DELETE FROM employee
WHERE phone = NULLABLE

QUESTION 25

You have a table named Customer that has columns named CustomerId, Firstname, and DateJoined. CustomerId is a primary key.

You execute the following query

```
SELECT CustomerID, FirstName, DateJoined
FROM Customer
```

How are the rows organized in the result set?

- A. In chronological order by DateJoined
- B. In the order in which the rows were inserted
- C. In no predictable order
- D. In alphabetical order by FirstName

QUESTION 26

You need to remove a view named EmployeeView from your database.

Which statement should you use ?

- a. DROP VIEW EmployeeView
- b. DELETE EmployeeView
- c. DELETE View EmployeeView
- d. DROP EmployeeView

QUESTION 27

Which SQL statement will return the country and number of orders in each country where the number of orders is less than 50?

OrderID represent an individual order, Country represent the country, and Orders represents the number of orders.

- A. SELECT COUNT (OrderId), Country
FROM Orders
HAVING COUNT(OrderId) < 50;
GROUP BY Country
- B. SELECT Country, OrderId
FROM Orders
GROUP BY Country
WHERE COUNT(OrderId) < 50;
- C. SELECT COUNT (OrderId), Country
FROM Orders
GROUP BY Country
HAVING COUNT(OrderId) < 50;
- D. SELECT Country, OrderId
FROM Orders
WHERE Count(OrderId) < 50;
GROUP BY Country

QUESTION 28

You have a table that contains the following data.

ProductID	ProductCategory
32	Books
25	Books
67	Movies
89	Movies

Which database term is used to describe the relationship between productid and productcategory.

- A. Deterministic
- B. Cohort
- C. Compositional
- D. Functionally dependent
- E. Relationally dependent

QUESTION 29

You execute a statement inside a transaction to delete 100 rows from a table. The transaction fails after only 40 rows are deleted.

What is the result in the database?

- A. The table will be corrupted
- B. 40 rows will be deleted from the table
- C. No rows will be deleted from the Table
- D. The transaction will restart

QUESTION 30

Your company stores customer social security numbers in a column named SSN in a table named customer. New compliance laws prohibit your company from storing this information. Running the query below causes an error to occur.

ALTER TABLE Customers

REMOVE SSN;

What changes are needed to the query above so that it removes the SSN column from the customers table ?

- A. ALTER TABLE Customers
DELETE SSN;
- B. ALTER TABLE Customers
DELETE COLUMN SSN;
- C. ALTER TABLE Customers
DROP SSN;
- D. ALTER TABLE Customers
DROP COLUMN SSN;

QUESTION 31

You have two tables named cars and colors as defined below . the two tables are related by colorid.

Cars table

Car Type	Car Model	Color ID
Sedan	2-Door	2
Truck	4-Door	NULL
Minivan	3-Door	1

Colors table

Color	Colorid
Red	1
gold	2

You run the following SQL statement

SELECT *

FROM Cars

LEFT OUTER JOIN Colors

ON Cars.ColorID = Color.ColorID

How many rows does the SQL statement return?

- A. 0
- B. 2
- C. 3
- D. 6

QUESTION 32

You are developing SQL statement to create a table.

Which two SQL keywords are valid to use in a create table statement? (Choose 2)

- A. Order By
- B. Primary Key
- C. Insert Into
- D. Constraint

QUESTION 33

The following table named Building stores data about building and their most recent inspection dates.

Field	Data Type	Allow Nulls
BuildingID	INT	FALSE
Address	VARCHAR(100)	FALSE
InspectorID	CHAR(3)	TRUE
InspectionDate	DATETIME	TRUE

The address field stores the building's address. The InspectionDate field stores the most recent inspection date. A value of null in the InspectionDate field means that the building has not yet been inspected.

You need to display the address of the earliest 10 buildings that have been inspected.

Complete the query by moving the appropriate keywords from the list on the left to the correct locations on the right. You may use each keyword once, more than once, or not at all.

Keywords: COUNT, GROUP BY 10, IS NOT NULL, IS NULL, ORDER BY, TOP 10, HAVING

Answer Area: SELECT TOP 10 Address FROM Building WHERE InspectionDate IS NOT NULL ORDER BY InspectionDate

QUESTION 34

Which query correctly returns a result set for all orders where the ship_state excludes Texas (TX) and Arizona (AZ)?

- A. SELECT * FROM Orders WHERE NOT Ship_State = 'TX' AND NOT Ship_State = 'AZ' ;
- B. SELECT * FROM Orders WHERE NOT Ship_State = 'TX' OR NOT Ship_State = 'AZ' ;
- C. SELECT * FROM Orders WHERE Ship_State NOT = 'TX' AND Ship_State NOT = 'AZ' ;
- D. SELECT * FROM Orders WHERE Ship_State NOT = 'TX' OR Ship_State NOT = 'AZ' ;

QUESTION 35

For each statement, select **true** or **false**.

Note : you will receive partial credit for each correct selection

	True	False
A full database backup is a copy of all of the data in the entire database	☒	☐
A transaction log backup backs up all the data in the database	☐	☒
A differential backup copies only data that was changed before the last full backup	☒	☒
A file or filegroup restore specifies a portion of the database to recover	☒	☐

QUESTION 36

To create a table named Games that contains the review scores of recently released video games.

You need to create a view that returns an alphabetical list of game names.

Name represents the game name.

Which query should you use?

- A. `CREATE VIEW Mygames As
SELECT Name FROM Games`
- B. `CREATE VIEW Mygames As
SELECT * FROM Games`
- C. `CREATE VIEW Mygames As
SELECT Name From Games
ORDER BY Name`
- D. `CREATE VIEW Mygames As
SELECT * From Games
Where Name BETWEEN 'A' AND 'Z'`

QUESTION 37

One reason to create a stored procedure is to :

- A. Minimized storage space
- B. **Improve performance**
- C. Give the user control of the query logic
- D. Bypass case sensitify requirements

QUESTION 38

You have a Department table and an Employee table in your database.

You need to ensure that an employee can be assigned to only an existing department.

What should you apply to the Employee table?

A. A foreign key

B. A primary key

C. An index

D. A unique constraint

E. A data type

QUESTION 39

You need to populate a table named `employeecopy` with data from an existing table named `employee`.

Which statement should you use?

A. `COPY * INTO Employee`

`SELECT *`

`FROM Employee`

B. `SELECT *`

`INTO Employeecopy`

`SELECT *`

`FROM Employee`

C. `INSERT *`

`FROM Employee`

`INTO Employeecopy`

D. `INSERT INTO Employeecopy`

`SELECT *`

`FROM Employee`

QUESTION 40

The `ItemsOnOrder` table contains the following data.

ID	ItemNumber	Quantity	UnitPrice	LineItemTotal
100	1	10	24.94	249.50
100	2	25	14.95	373.75
100	3	25	19.99	499.75
101	4	10	25.00	250.00
102	5	10	29.99	299.00

You need to create a query that displays the total number of orders, the average line item total, the highest line item total, and the grand total of all items ordered.

`LineItemTotal` represents the line item total.

Which query should you use?

A. `SELECT COUNT (ID), AVG(LineItemTotal), MAX(LineItemTotal),`

`SUM(LineItemTotal)`

`FROM ItemsOnOrder`

`HAVING ItemNumber, Quantity, UnitPrice`

- B. SELECT COUNT (ID), AVG(Unitprice+Quantity), MAX(UnitPrice+Quantity), SUM(UnitPrice+Quantity)
FROM ItemOnOrder
GROUP By ItemNumber, LineItemTotal
- C. SELECT SUM(ID), AVG(LineItemTotal), MAX(LineItemTotal), SUM(LineItemTotal)
FROM ItemOnOrder
- D. SELECT COUNT(ID), AVG(LineItemTotal), MAX(LineItemTotal), SUM
(LineItemTotal) FROM ItemOnOrder

QUESTION 41

Your database contains a table named Customer. You need to delete the record from the Customer table that has a CustomerID of 12345.

Which statement should you use?

- A. DELETE CustomerID
FORM Customer
WHERE CustomerID = 12345
- B. DELETE FROM Customer
WHERE CustomerID = 12345
- C. UPDATE Customer
DELETE *
WHERE CustomerID = 12345
- D. UPDATE CustomerID
FROM Customer
DELETE *
WHERE CustomerID = 12345

QUESTION 42

You have a table named Product that contains the following data.

ProductID	ProductName	CategoryID
3296	Spoon	2222
1114	Chair	4444

The ProductID column is the primary key. The CategoryID column is a foreign key to a separate table named Category.

You execute the following statement.

INSERT INTO Product
VALUES (3296, 'Table', 4444)

What is the result?

- A. A new row in the Product table
- B. A syntax error
- C. A foreign key constraint violation
- D. A new row in the Category table
- E. A primary key constraint violation

QUESTION 43

You delete rows in a table named Order. The corresponding rows in the OrderItem table are automatically deleted.

This process is an example of a/an:

- A. waterfall delete
- B. inherited delete
- C. cascade delete
- D. domino delete
- E. functional delete

QUESTION 44

Which keyword combines the results of two queries and returns only rows that appear in both result sets?

- A. UNION
- B. INTERSECT
- C. ALL
- D. JOIN

QUESTION 45

You create the following table named LoanedBooks, which lists how many books you have on loan to your friends.

ID	Name	City	Books
1	Harry	New York	2
2	Sally	San Fransisco	3
3	Sue	Chicago	0
4	Harry	San Fransisco	3
5	William	Detroit	1

Harry in San Fransisco returns your books.

Which statement will correctly update your table?

- A. UPDATE LoanedBooks
SET Books = 0
WHERE (Name = 'Harry' OR City = 'San Fransisco')
- B. UPDATE LoanedBooks
SET Books = 0
WHERE (Name In 'Harry', 'San Fransisco')
- C. UPDATE LoanedBooks
SET Books = 0
WHERE (Name 'Harry' AND City = 'San Fransisco')
- D. INSERT INTO LoanedBooks
SET Books = 0
WHERE ID = 4

QUESTION 46

You have a table named Product. You create a view that includes all the products from the Product table that are in the Furniture category.

You execute a statement on the Product table that deletes all the products in the Furniture category.

After you execute the statement, the result set of the view is:

- A. empty
- B. deleted
- C. unchanged
- D. archived

QUESTION 47

You have two tables. Each table has three rows. You create a SQL query that uses a cross join. The query does not include a WHERE clause.

How many rows will be included in the Cartesian product of these two tables?

- A. 0
- B. 3
- C. 6
- D. 9

QUESTION 48

You have a table named Customer. You need to create a new column named District. Which statement should you use?

- A. ALTER TABLE Customer
MODIFY (District INTEGER)
- B. ALTER TABLE Customer
ADD (INTEGER District)
- C. ALTER TABLE Customer
ADD (District INTEGER)
- D. MODIFY TABLE Customer
ADD (INTEGER District)

QUESTION 49

One difference between a function and a stored procedure is that a function.

- A. cannot accept parameters
- B. cannot contain a transaction
- C. must be called from a trigger
- D. must return a value

QUESTION 50

Which SQL statement is a data manipulation language statement?

- A. ALTER TABLE Employee

ADD EmployeeName Varchar;

B. SELECT EmployeeName FROM Employee

WHERE EmployeeName = 'Jack Smith';

C. SELECT * INTO Employee FROM NewHires

INSERT INTO Employee

D. VALUES ('Jack Smith');

QUESTION 51

You create the following table, which displays winners of past pet shows.

Pet Competition Winners

ID	Event	Year	PetID	Birthdate	City
1	Cutest Pet Fest	2009	RG459	June 23, 2008	Chicago
2	Pet Babies Invitational	2010	SP372	August 12, 2009	Montreal
3	Adorable Friends	2011	BD116	May 1, 2010	Atlanta
4	AWHH Fest 2012	2012	FD228	August 12, 2009	Detroit

Which column prevents the table from being in third normal form?

A. PetID

B. Birthdate

C. ID

D. Year

QUESTION 52

For each statement, select **True** or **False**.

Note: You will receive partial credit for each correct selection.

	True	False
You can delete data by using a stored procedure.	☒	☐
A function must have a return value.	☒	☐
A stored procedure must have a return value.	☐	☒

QUESTION 53

You are creating a database object named Student to store the following data:

ID	Name	Age
1	Rene	18
2	Tia	22
3	Oliver	25

Which syntax should you use to create the object?

A. CREATE TABLE Student(

ID INT,

Name VARCHAR (100),

Age INT)

- B. CREATE Student(
ID INT,
Name VARCHAR (100),
Age INT)
- C. CREATE TABLE (
ID INT,
Name VARCHAR (100),
Age INT)
- D. CREATE (
TABLE Student
ID INT,
Name VARCHAR (100),

QUESTION 54

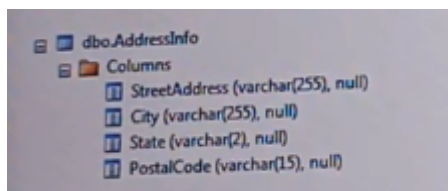
You need to disable User1's access to view the data in the Customer table.

Which statement should you use?

- A. REVOKE User1
FROM Customer
- B. REVOKE SELECT ON Customer
FROM User1
- C. REMOVE SELECT ON Customer
FROM User1
- D. REMOVE User1
FROM Customer

QUESTION 55

You have a database table that contains the following columns:



You must insert the following record into the table:

StreetAddress	City	State	PostalCode
1234 main Street	Dallas	TX	75201

Which two SQL statement can you use? (Choose 2)

- A. INSERT INTO AddressInfo ('1234 Main Street', 'Dallas', 'TX', '75201')
VALUES ([StreetAddress], [City], [State], [PostalCode])
- B. INSERT INTO AddressInfo VALUES ('1234 Main Street', 'Dallas', 'TX', '75201')
- C. UPDATE AddressInfo SET [StreetAddress] = '123 Main Street', [City] = 'Dallas', [State] = 'TX', [PostalCode] = '75201'

D. **INSERT INTO AddressInfo ([StreetAddress], [City], [State], [PostalCode])
VALUES ('1234 Main Street', 'Dallas', 'TX', '75201')**

QUESTION 56

You're the database admin for a university that stores student information in a table named `student`. This table has a foreign key named `courseID` that links to the primary key in a table named `courses`.

You need to return the student name and course name using both tables where there is a match on `courseID`.

You execute the following query:

```
SELECT students.name, courses.name FROM students  
OUTER JOIN courses WHERE student.courseID = courses.courseID
```

The query returns an error.

How should you correct the query?

- A. `SELECT students.name, courses.name FROM students
INNER JOIN courses WHERE students.courseID = courses.courseID`
- B. **`SELECT students.name, courses.name FROM students
INNER JOIN courses ON students.courseID = courses.courseID`**
- C. `SELECT students.name, courses.name FROM students
INNER JOIN ON students.courseID = courses.courseID`
- D. `SELECT students.name, courses.name FROM students
INNER JOIN On courses
WHERE students.courseID = courses.courseID`

QUESTION 57

Another developer is trying to add a column named `Prefix` in a table by using the following query:

```
ALTER TABLE Person ADD Prefix varchar(4) NOT NULL;
```

The developer receives an error message after running the query. The developer is unable to remember the exact error message.

What is likely the cause of this problem?

- A. The `Person` table is empty.
- B. You should run the `DROP CONSTRAINT` query before running the `ALTER TABLE` query.
- C. **The `varchar(4)` data types is invalid for the `Person` table.**
- D. **The `DEFAULT` keyword should be used to specify a default value.**

QUESTION 58

You have a table named `Student` that contains 100 rows. Some of the rows have a `NULL` value in the `FirstName` column.

You execute the following statement:

```
DELETE FROM Student
```

What is the result?

- A. All rows and the table definition will be deleted
- B. All rows containing a NULL value in the FirstName column will be deleted
- C. All rows in the table will be deleted
- D. You will receive an error message

QUESTION 59

You have a table that contains the following data.

ProductID	ProductCategory
32	Books
25	Books
67	Movies
89	Movies

Which database term is used to describe the relationship between ProductID and ProductCategory?

- A. compositional
- B. cohort
- C. deterministic
- D. relationally dependent
- E. functionally dependent

QUESTION 60

You have two tables named SalesPerson and Sales.

You need to ensure that each record in the Sales table has a valid associated salesperson record in the SalesPerson table.

Which database object should you add to the Sales table?

- A. Clustered index
- B. Primary key
- C. Foreign key
- D. Nonclustered index

QUESTION 61

You have a table named Employee that includes the following columns:

EmployeeID

EmployeeName

Which statement should you use to return the number of rows in the table?

- A. SELECT *
FROM Employee
- B. SELECT COUNT(*)
FROM Employee

- C. SELECT COUNT(rows)
FROM Employee
- D. SELECT SUM(*)
FROM Employee

QUESTION 62

The products table contains the following data.

ItemNumber	ItemName	ItemDescription	Price
1	Bonbon Box	Chocolate Truffles, Black Forest Truffles, and Cherry Bonbons	24.95
2	Brownie Mini Bites	Caramel Nut, German Chocolate, Chocolate Swirl, and Double Dutch Brownies	14.95
3	Cappuccino Companion Collection	Tasty treats to accompany your favorite java, Includes Strawberry Thumbprints, Chocoate Chip Cookies, and Biscotti.	21.50
4	Citrus Cooler Gift Box	Refreshing citrus cookies, including Orange Chiffon Meltaways, Key Lime Bars, Lemon Coolers, and Tangerine Coconut Smoothies	19.99
5	Fruit Jewels Assortment	Fruity Favorites, including Apricot Jewels, Cherry Bonbons, Coconut Patties, and Strawberry Thumbprints	29.99

Which query will retrieve ItemName and Price when chocolate appears in the ItemDescription column?

- A. SELECT ItemName, Price
FROM Products
Where ItemDescription = 'chocolate';
- B. SELECT ItemName, Price
FROM Products
WHERE ItemDescription LIKE '%chocolate%';
- C. SELECT ItemName, Price
FROM Products
WHERE ItemDescription LIKE 'chocolate';
- D. SELECT ItemName, Price
FROM Products
WHERE ItemDescription IN '%chocolate%';

QUESTION 63

You need to combine the results of two queries into a single result that contains all of the rows from both queries.

Which SQL statement should you use?

- A. UNION

- B. JOIN
- C. EXCEPT
- D. TRUNCATE

QUESTION 65

You need to set up a database to provide a view of North American mammals (NorthAmericanMammals_View).

Move the appropriate code segments from the list on the left to the correct locations on the right. You may use each code segment once, more than once, or not at all.

Note: You will receive partial credit for each correct response.

Code Segments	Answer Area
CREATE VIEW [dbo].[NorthAmericanMammals_View]	CREATE VIEW [dbo].[NorthAmericanMammals_View]
INSERT VIEW [dbo].[NorthAmericanMammals_View]	AS SELECT a.Id, a.Name
AS JOIN a.Id, a.Name	FROM Animal a
AS SELECT a.Id, a.Name	WHERE a.Class = 'Mammals'
FROM Animal a	AND a.InNorthAmerica = 1
JOIN Animal a	

QUESTION 66

Which statement creates a composite key?

- A. CREATE TABLE Order
(OrderID INTEGER,
OrderItemID INTEGER,
PRIMARY KEY OrderID,
PRIMARY KEY OrderItemID)
- B. CREATE TABLE Order
(OrderID INTEGER PRIMARY KEY,
OrderItemID INTEGER PRIMARY KEY)
- C. CREATE TABLE Order
(OrderID INTEGER,
OrderItemID INTEGER,
PRIMARY KEY)
- D. CREATE TABLE Order
(OrderID INTEGER,
OrderItemID INTEGER,
PRIMARY KEY(OrderID, OrderItemID))

QUESTION 67

A table has a clustered index.

Complete the statements by selecting the correct option from each drop-down list.

Note: You will receive partial credit for each correct selection.

Answer Area

A clustered index improves the performance of queries that:

A clustered index improves the performance of queries on columns that:

QUESTION 68

You work at a coffee shop. They ask you to set up a website that stores charges on purchases. You need to recommend a data type in a database table to run financial functions against the charged amounts.

Which data type should you recommend?

- A. Decimal
- B. Binary
- C. Bit
- D. Varchar

QUESTION 69

Which statement removes all rows from a table without logging the individual row deletions?

- A. ALTER TABLE
- B. DROP TABLE
- C. TRUNCATE TABLE
- D. CREATE TABLE

QUESTION 70

A database table named Cars is defined as shown.

Type	Color	Origin
Sedan	Red	Japan
Truck	Red	USA
Minivan	Silver	Japan
Hatchback	Red	Japan
Compact	Black	Japan
SUV	Silver	Gemany
Convertible	Black	USA
Hybrid	Black	Germany

You have the following SQL statement:

SELECT *

FROM Cars C

WHERE c.Origin <> 'USA' AND c.Color <> 'Black'

How many rows are returned by the SQL statement?

- A. 4
- B. 5
- C. 6
- D. 7

QUESTION 71

You execute the following query:

```
SELECT EmployeeID, FirstName, DepartmentName  
FROM Employee, Department
```

This type of operation is called a/an:

- A. Equi-join
- B. Cartesian product
- C. Intersection
- D. Outer join

QUESTION 72

A table named Flight contains the following fields:

Field	Data Type	Allow Nulls
ID	INT	FALSE
FlightNumber	VARCHAR(8)	FALSE
OriginAirport	CHAR(3)	FALSE
DestinationAirport	CHAR(3)	FALSE
DepartureTime	DATETIME	FALSE
ArrivalTime	DATETIME	FALSE

You need to display the flight numbers of all flights that arrive at LaGuardia Airport (LGA) later today. The results should be sorted by the most recent arrival time.

Which query should you use? Complete the code by selecting the correct option from each drop-down list.

Note: You will receive partial credit for each correct selection.

```
sql                                                                    Salin Edit

SELECT FlightNumber
FROM Flight
WHERE DestinationAirport = 'LGA'
      AND ArrivalTime > GETDATE()
ORDER BY ArrivalTime DESC;
```

Answer Area

```
SELECT FlightNumber FROM Flight
WHERE DestinationAirport = 'LGA' AND
ArrivalTime = GETDATE()
ORDER BY ArrivalTime ASC
```

QUESTION 73

Which query returns a result set of orders placed after January 2017 in all states except California (CA)?

- A. `SELECT * FROM orders WHERE order_date > '2017-01-01' AND ship_state <> 'CA'`
- B. `SELECT * FROM orders WHERE order_date > '2017-01-01' OR ship_state <> 'CA'`
- C. `SELECT * FROM orders WHERE order_date ? '2017-01-01' AND ship_state LIKE 'CA'`
- D. `SELECT * FROM orders WHERE order_date > '2017-01-01' OR ship_state LIKE 'CA'`

QUESTION 74

The Customers tables is defined as follows:

CustomerID	Address	Region	Country
123	1234 Main Street	Nevada	United States
456	456 Rue de Paris	Paris	France
789	789 Elm Street	Maine	United States

You need to write a SQL query that will return the number of customers in each country that has fewer than 50 customers

Which query should you use?

- A. `SELECT Country, CustomerID`
`FROM Customers`
`GROUP BY Country`
`WHERE COUNT (CustomerID) < 50;`
- B. `Select COUNT(CustomerID), Country`
`FROM CUSTOMERS`
`HAVING COUNT(CustomerID) < 50`
`GROUP BY Country;`
- C. `SELECT Country, CustomerID`
`FROM Customers`
`WHERE COUNT(CustomerID) < 50`
`GROUP BY Country;`
- D. `SELECT COUNT(CustomerID), Country`
`FROM Customers`
`GROUP BY Country`
`HAVING COUNT(CustomerID) < 50;`

QUESTION 75

You need to normalize a database to first normal form
Which two requirements must you meet? (Choose2.)

- A. Exclude duplicate rows
- B. Exclude repeating groups
- C. Exclude foreign keys
- D. Exclude composite keys

QUESTION 76

ID	Name	State	Type	Description	Date
1000	Healthy Eats	NY	ABA	Delivers meal kits	02/01/1988
1001	Rock Music Zone	UT	AZZ	Sells musical instruments	04/10/2022
1002	Sunset Web	UT	CYY	Builds custom software solutions	08/01/1943
1003	CompanyPro	CA	MCC	Create custom learning solutions	12/31/2016

You want to create a view that retrieves only the ID, Name and Type from the table for all companies in the state of UT

How should you construct the query?

Complete the markup by selecting the correct option from each drop-down list.

Note: You will receive partial credit for each correct selection.

Answer Area

QUESTION 77

What is a benefit of creating a stored procedure?

- A. Gives the user control of the query logic
- B. Improves performance
- C. Minimizes storage space
- D. Bypasses case sensitivity requirements

QUESTION 78

Which keyword combines the results of two queries and returns only rows that appear in both result sets?

- A. UNION
- B. **INTERSECT**
- C. ALL
- D. JOIN

QUESTION 79

Which SQL Statement is a data definition language (DDL)

- A. SELECT * INTO Employee FROM NewHires;
- B. SELECT EmployeeName FROM Employee
WHERE EmployeeName = 'Jack Smith';
- C. **ALTER TABLE Employee
ADD EmployeeName Varchar;**
- D. INSERT INTO Employee
VALUES ('Jack Smith');

QUESTION 80

You run the following two queries:

```
SELECT [Machine].[Name] AS Machine FROM [Machine]  
JOIN [Assignment] ON MachineID = Machine.ID;
```

```
SELECT [Operator].[Name] AS Operator FROM [Operator]  
LEFT JOIN [Assignment] ON OperatorID = Operator.ID;
```

Evaluate the graphic and answer each question by entering the correct number in each spin box.

Note: You will receive credit for each correct selection

Machine

ID	Name	Model
1	Packing Machine	PK10
2	Painting Machine	PM02

3	Wrapping Machine	WR03
---	------------------	------

ID	Name	Experience
1	Barry P	1
2	Heidi S	1
3	Nina V	2

ID	MachineID	OperatorID
1	1	1
2	2	2

Evaluate the queries and the data tables and answer the questions in the answer area.

Note: You will receive partial credit for each correct selection

Answer Area

How many rows are returned by the first query?

How many rows are returned by the second query?

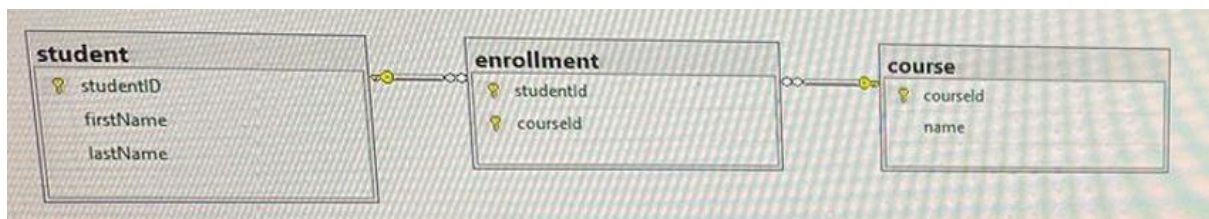
QUESTION 81

How is a function different from a stored procedure?

- A. A function must be called from a trigger
- B. A function must return a value
- C. A function cannot accept parameters
- D. A function cannot contain a transaction

QUESTION 82

You have the following tables:



You need to return the student name and each course they are taking. Student who are not enrolled should not be returned

You execute the following query

```
SELECT student.firstname, student.lastname, course.name FROM Student  
OUTER JOIN course WHERE student.courseID = course.courseID;
```

The query return an error

How should you correct the query?

- A. `SELECT student.firstname, student.lastname, course.name FROM course
INNER JOIN enrollment ON enrollment.courseID = course.courseID
INNER JOIN student ON enrollment.studentID = student.studentID;`
- B. `SELECT firstname, lastname, name FROM student
INNER JOIN ON course, enrollment
WHERE enrollment.courseID = course.courseID AND
enrollment.studentID = student.studentID;`
- C. `SELECT student.firstname, student.lastname, course.name FROM
student
INNER JOIN ON course, enrollment
WHERE enrollment.courseID = course.courseID AND
enrollment.student ID = student.studentID;`
- D. `SELECT firstname, lastname, name FROM course
INNER JOIN enrollment WHERE enrollment.courseID =
course.courseID
INNER JOIN student WHERE enrollment.studentID = student.studentID;`

QUESTION 83

You are designing a database that must meet these requirements:

- A table name MACHINE represents all the machines in a factory. The ID uniquely identifies each machine.
- A table named OPERATOR represents all the operators in a factory. The ID field uniquely identifies each operator
- Each operator can be assigned to zero or more machines
- Each machine can be assigned to zero or more operators

You need to design the entity relationship diagram (ERD) to meet these requirements and enforce referential integrity.

Select the correct keys in the dialog box

Answer Area

Machine
ID

Name
Model

Operator
ID
Name
Experience

Assignment
ID
Name
Model

QUESTION 84

You accept an IT internship at a local charity. The charity asks you to keep a record of its volunteers by using a database table named Volunteer. The table has the following columns and rows:

Id	GivenName
1	Tia
2	Susana
3	Joey

When volunteer information changes, you must update the table. You need to change Tia's name to Kimberly.

Which statement should you choose?

- SET Volunteer = 'Kimberly'
WHERE GivenName = 'Tia';
- UPDATE Volunteer
SET GivenName = "Kimberly"
WHERE GivenName = 'Tia';
- UPDATE Givenilame = 'Kimberly'
FROM Volunteer
WHERE Givenllame = 'Tia';
- SET Giventlame = 'Kimberly'
FROM Volunteer
WHERE Givenlame = 'Tia';